

AKS-managed Azure Active Directory/Microsoft Entra integration

- AKS provides three authentication and authorization options:

Local accounts with Kubernetes RBAC

Use built-in Kubernetes role-based access control for authorization checks on the cluster.

Azure AD authentication with Kubernetes RBAC

Use Azure AD for authentication and Kubernetes native RBAC for authorization.

Azure AD authentication with Azure RBAC

Use Azure role assignments for authorization checks on the cluster.

Source: Azure Portal

Local accounts with Kubernetes RBAC

- By default, an AKS cluster is created with local accounts using Kubernetes RBAC, with no integration between AKS and Azure AD
 - You can use native Kubernetes options like client certificates or bearer tokens
 - “az aks get-credentials” but everyone will have full access

AKS-managed AAD integration

- Irreversible
- OpenID Connect is used to enable authentication
- Azure AD Premium offer conditional policies to restrict access further, e.g., by location or JIT
- Integrating AKS cluster with Azure AD allows fine-grained control over user access using Azure RBAC and Kubernetes RBAC

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Azure RBAC

- Uses built-in (for general purposes) or custom Azure roles assigned to Azure AD users, groups, or service principals at the AKS cluster or Kubernetes object level.
- Example of built-in role:
 - Azure Kubernetes Service Cluster User Role. Contains actions:
 - Microsoft.ContainerService/managedClusters/read
 - Microsoft.ContainerService/managedClusters/listClusterUserCredential/action
- Examples of actions that can be used for custom roles:
 - AKS cluster level:
 - Microsoft.ContainerService/managedClusters/stop/action
 - Microsoft.ContainerService/managedClusters/start/action
 - Kubernetes level:
 - Microsoft.ContainerService/managedClusters/apps/deployments/read
 - Microsoft.ContainerService/managedClusters/apps/deployments/write
 - Microsoft.ContainerService/managedClusters/apps/deployments/delete
- Built-in and custom roles can be assigned at the Azure level

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Kubernetes RBAC

- About permissions at the Kubernetes level only and requires one or more groups to be designated as administrators
 - Will create ClusterRoleBinding for that group(s) for the default ClusterRole cluster-admin
- You can make use of Roles, ClusterRoles, RoleBindings and ClusterRoleBindings to control what specific AAD users or AAD groups can see or do
- Roles and RoleBindings apply at namespace level, while ClusterRoles and ClusterRoleBindings are at the cluster level

Note on Local accounts

- Even with AAD integration and Azure RBAC/Kubernetes RBAC, local accounts can still be enabled. You can disable this or have it enabled based on your considerations